

WMO State of the Global Climate 2023-2024

Published: April 2024 | World Meteorological Organization (WMO)

1. Global Climate Overview

The World Meteorological Organization consolidates observational data from international meteorological and hydrological services to track critical global climate variables.

2. Atmospheric Composition

WMO consolidated observations confirmed atmospheric CO₂ concentrations reached 421.5 ppm in 2023. For the prior observation year 2022, global atmospheric carbon dioxide stood at 418.7 ppm across the Global Atmosphere Watch network.

3. Surface Temperature Records

The global average surface temperature anomaly was recorded at 1.45 °C above pre-industrial levels for the calendar year 2023. In comparison, the global mean temperature anomaly for 2022 was 1.15 °C.

4. Cryosphere Indicators

Antarctic sea ice maximum extent was observed at 16.96 million sq km in September 2023, confirming unprecedented reductions compared to the long-term climatological median of 18.71 million sq km.

5. Renewable Power Additions

Global additions of renewable power capacity totaled 510 GW in 2023, representing a sharp acceleration from renewable capacity additions of 381 GW in 2022.